

3B

April 16, 2026

```
[ ]: import tensorflow as tf  
print(tf.__version__)
```

```
[2]: import sys  
! "{sys.executable}" -m pip install matplotlib
```

```
Defaulting to user installation because normal site-packages is not writeable  
Collecting matplotlib  
  Downloading matplotlib-3.9.4-cp39-cp39-win_amd64.whl (7.8 MB)  
    ----- 7.8/7.8 MB 555.6 kB/s eta 0:00:00  
Collecting fonttools>=4.22.0  
  Downloading fonttools-4.60.2-cp39-cp39-win_amd64.whl (1.5 MB)  
    ----- 1.5/1.5 MB 984.4 kB/s eta 0:00:00  
Requirement already satisfied: packaging>=20.0 in  
c:\users\lenovo\appdata\roaming\python\python39\site-packages (from matplotlib  
(26.0))  
Collecting importlib-resources>=3.2.0  
  Downloading importlib_resources-6.5.2-py3-none-any.whl (37 kB)  
Collecting cycler>=0.10  
  Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)  
Requirement already satisfied: pillow>=8 in  
c:\users\lenovo\appdata\roaming\python\python39\site-packages (from matplotlib  
(11.3.0))  
Collecting contourpy>=1.0.1  
  Downloading contourpy-1.3.0-cp39-cp39-win_amd64.whl (211 kB)  
    ----- 211.8/211.8 KB 1.1 MB/s eta 0:00:00  
Requirement already satisfied: python-dateutil>=2.7 in  
c:\users\lenovo\appdata\roaming\python\python39\site-packages (from matplotlib  
(2.9.0.post0))  
Requirement already satisfied: numpy>=1.23 in  
c:\users\lenovo\appdata\roaming\python\python39\site-packages (from matplotlib  
(2.0.2))  
Collecting kiwisolver>=1.3.1  
  Downloading kiwisolver-1.4.7-cp39-cp39-win_amd64.whl (55 kB)  
    ----- 55.8/55.8 KB 1.4 MB/s eta 0:00:00  
Collecting pyparsing>=2.3.1  
  Downloading pyparsing-3.3.2-py3-none-any.whl (122 kB)  
    ----- 122.8/122.8 KB 554.4 kB/s eta 0:00:00
```

Requirement already satisfied: zipp>=3.1.0 in
c:\users\lenovo\appdata\roaming\python\python39\site-packages (from importlib-
resources>=3.2.0->matplotlib) (3.23.0)
Requirement already satisfied: six>=1.5 in
c:\users\lenovo\appdata\roaming\python\python39\site-packages (from python-
dateutil>=2.7->matplotlib) (1.17.0)
Installing collected packages: pyparsing, kiwisolver, importlib-resources,
fonttools, cycycler, contourpy, matplotlib
Successfully installed contourpy-1.3.0 cycycler-0.12.1 fonttools-4.60.2 importlib-
resources-6.5.2 kiwisolver-1.4.7 matplotlib-3.9.4 pyparsing-3.3.2

WARNING: The scripts fonttools.exe, pyftmerge.exe, pyftsubset.exe and ttx.exe
are installed in 'C:\Users\Lenovo\AppData\Roaming\Python\Python39\Scripts' which
is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this
warning, use --no-warn-script-location.

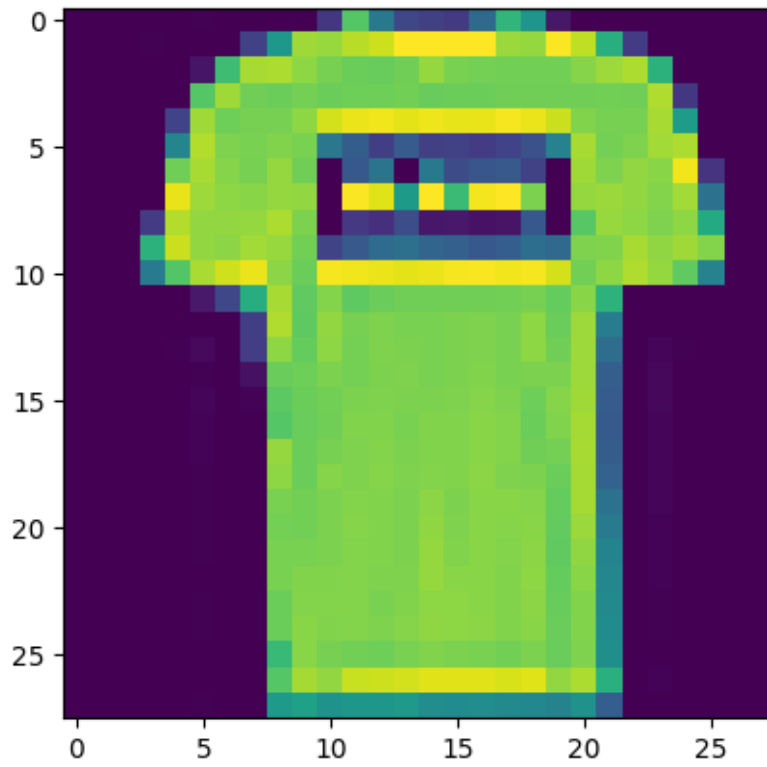
WARNING: You are using pip version 22.0.4; however, version 26.0.1 is available.
You should consider upgrading via the 'C:\Program Files (x86)\Microsoft Visual
Studio\Shared\Python39_64\python.exe -m pip install --upgrade pip' command.

```
[3]: import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
import numpy as np
```

```
[4]: (x_train, y_train), (x_test, y_test) = keras.datasets.fashion_mnist.load_data()
```

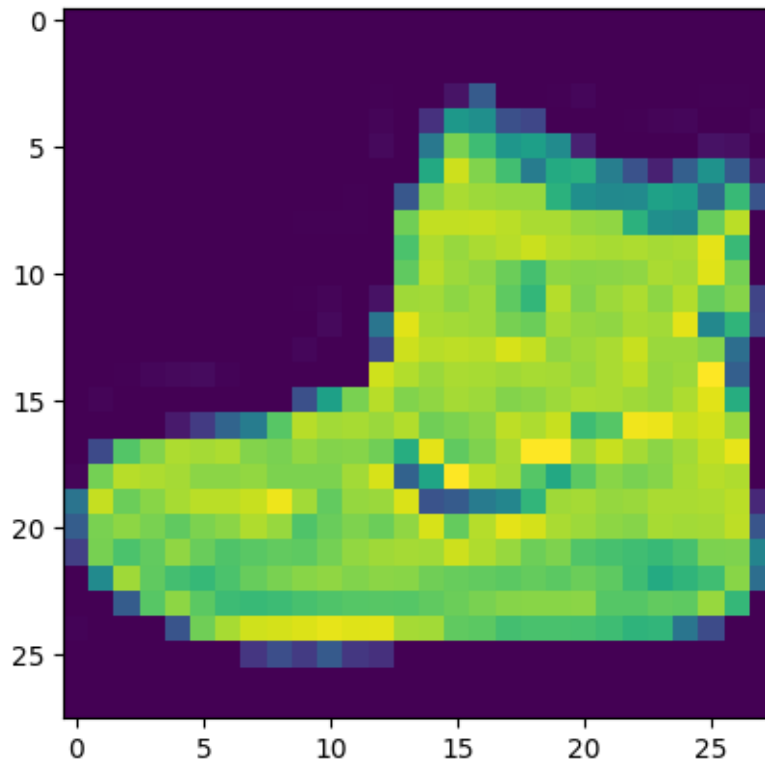
Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx1-ubyte.gz>
29515/29515 0s 3us/step
Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-images-idx3-ubyte.gz>
26421880/26421880 41s
2us/step
Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-labels-idx1-ubyte.gz>
5148/5148 0s 2us/step
Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/t10k-images-idx3-ubyte.gz>
4422102/4422102 6s
1us/step

```
[5]: plt.imshow(x_train[1])
plt.show()
```



```
[6]: plt.imshow(x_train[0])
```

```
[6]: <matplotlib.image.AxesImage at 0x252c14e3b20>
```



```
[7]: x_train = x_train.astype('float32') / 255.0  
     x_test = x_test.astype('float32') / 255.0
```

```
[8]: x_train = x_train.reshape(-1, 28, 28, 1)  
     x_test = x_test.reshape(-1, 28, 28, 1)
```

```
[9]: x_train.shape
```

```
[9]: (60000, 28, 28, 1)
```

```
[10]: x_test.shape
```

```
[10]: (10000, 28, 28, 1)
```

```
[11]: y_train.shape
```

```
[11]: (60000,)
```

```
[12]: y_test.shape
```

```
[12]: (10000,)
```

```
[17]: model = keras.Sequential([
    keras.Input(shape=(28,28,1)),    # Add this

    keras.layers.Conv2D(32, (3,3), activation='relu'),
    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),

    keras.layers.Conv2D(64, (3,3), activation='relu'),
    keras.layers.MaxPooling2D((2,2)),
    keras.layers.Dropout(0.25),

    keras.layers.Conv2D(128, (3,3), activation='relu'),

    keras.layers.Flatten(),

    keras.layers.Dense(128, activation='relu'),
    keras.layers.Dropout(0.25),

    keras.layers.Dense(10, activation='softmax')
])
```

```
[18]: model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	
↳ Param #		
conv2d_3 (Conv2D)	(None, 26, 26, 32)	↳
↳ 320		
max_pooling2d_2 (MaxPooling2D)	(None, 13, 13, 32)	↳
↳ 0		
dropout_3 (Dropout)	(None, 13, 13, 32)	↳
↳ 0		
conv2d_4 (Conv2D)	(None, 11, 11, 64)	↳
↳ 18,496		
max_pooling2d_3 (MaxPooling2D)	(None, 5, 5, 64)	↳
↳ 0		
dropout_4 (Dropout)	(None, 5, 5, 64)	↳
↳ 0		

```

conv2d_5 (Conv2D)                                (None, 3, 3, 128)
↳73,856
flatten_1 (Flatten)                              (None, 1152)
↳ 0
dense_2 (Dense)                                  (None, 128)
↳147,584
dropout_5 (Dropout)                             (None, 128)
↳ 0
dense_3 (Dense)                                  (None, 10)
↳1,290

```

Total params: 241,546 (943.54 KB)

Trainable params: 241,546 (943.54 KB)

Non-trainable params: 0 (0.00 B)

```
[19]: model.compile(optimizer='adam', loss='sparse_categorical_crossentropy',
↳metrics=['accuracy'])
```

```
[20]: history = model.fit(x_train, y_train, epochs=10, validation_data=(x_test,
↳y_test))
```

```

Epoch 1/10
1875/1875      80s 39ms/step -
accuracy: 0.7100 - loss: 0.7806 - val_accuracy: 0.8616 - val_loss: 0.3740
Epoch 2/10
1875/1875      88s 42ms/step -
accuracy: 0.8590 - loss: 0.3880 - val_accuracy: 0.8854 - val_loss: 0.3140
Epoch 3/10
1875/1875      83s 43ms/step -
accuracy: 0.8793 - loss: 0.3283 - val_accuracy: 0.8916 - val_loss: 0.2977
Epoch 4/10
1875/1875      81s 43ms/step -
accuracy: 0.8910 - loss: 0.2982 - val_accuracy: 0.8984 - val_loss: 0.2843
Epoch 5/10
1875/1875      83s 43ms/step -
accuracy: 0.8992 - loss: 0.2737 - val_accuracy: 0.9027 - val_loss: 0.2682
Epoch 6/10
1875/1875      76s 40ms/step -

```

```
accuracy: 0.9022 - loss: 0.2616 - val_accuracy: 0.9002 - val_loss: 0.2658
Epoch 7/10
1875/1875          74s 39ms/step -
accuracy: 0.9076 - loss: 0.2539 - val_accuracy: 0.8937 - val_loss: 0.2847
Epoch 8/10
1875/1875          79s 42ms/step -
accuracy: 0.9082 - loss: 0.2437 - val_accuracy: 0.9054 - val_loss: 0.2608
Epoch 9/10
1875/1875          83s 42ms/step -
accuracy: 0.9114 - loss: 0.2325 - val_accuracy: 0.9028 - val_loss: 0.2630
Epoch 10/10
1875/1875          79s 41ms/step -
accuracy: 0.9148 - loss: 0.2247 - val_accuracy: 0.9123 - val_loss: 0.2433
```

```
[22]: print('Test accuracy:', test_acc)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[22], line 1
----> 1 print('Test accuracy:', test_acc)

NameError: name 'test_acc' is not defined
```

```
[ ]:
```